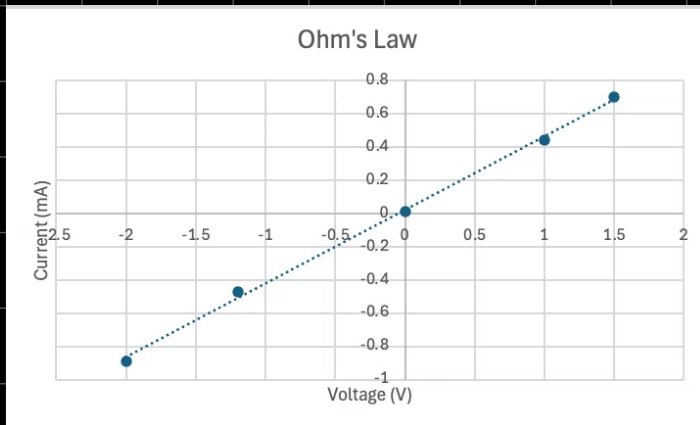


1)



$$\text{slope} = \frac{1\text{V}}{0.44\text{ mA}} \approx \frac{2.27\text{ V}}{\text{mA}}$$

$$R = 2.27 \times 10^3 = \underline{2.27\text{ k}\Omega}$$

2) $V = IR \rightarrow R = \frac{V}{I} = \frac{120\text{V}}{10\text{A}} = 12\ \Omega$

3a) $P_{\text{abs}} = IV = 1\text{ pA} \cdot 6\text{V} = 6\text{ pW}$

3b) $P_{\text{abs}} = IV = 10\text{ mA} \cdot 1\text{V} = 10\text{ mW}$

3c) $P_{\text{abs}} = -I_{\text{entering}} V = -(2\text{ A} \cdot 10\text{V}) = -20\text{W}$

4) 2V: $P = IV = -2\text{ A} \cdot 2\text{V} = -4\text{W}$

8V: $P = IV = -2\text{ A} \cdot 8\text{V} = -16\text{W}$

4A: $P = IV = 4\text{ A} \cdot 10\text{V} = +40\text{W}$

5A: $P = IV = 5\text{ A} \cdot 10\text{V} = -50\text{W}$

$$3A; P = IV = 3A \cdot 10V = +30W$$

$$P_d = P_A$$

$$4 + 16 + 30 = 30 + 40$$

$$70 = 70$$

$$5) \quad 1 \text{ m}^2; 1000 \text{ W/m}^2 \times 0.15 = 150 \text{ W/m}^2$$

$$\text{Area for } 1500 \text{ W: } \frac{1500 \text{ W}}{150 \text{ W/m}^2} = 10 \text{ m}^2$$

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$$a) \quad P = 2V \cdot 4A = 8W \text{ Delivering}$$

$$b) \quad P = 3V \cdot -2A = -6W \text{ Absorbing}$$

$$c) \quad P = -6V \cdot -8A = 48W \text{ Absorbing}$$

$$d) \quad P = -5V \cdot 1A = -5W \text{ Absorbing}$$

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$$R_{eq} = R_1 + R_2$$

$$I = \frac{V_s}{R_1 + R_2}$$

$$P_s = V_s \cdot I = V_s \cdot$$

$$P_{abs} = I^2 R_1 + I^2 R_2$$

$$V_{R_2} = I \cdot R_2 = \left(\frac{V_s}{R_1 + R_2} \right) R_2 = V_s \frac{R_2}{R_1 + R_2}$$